

$$y = f(x)$$

$$\frac{\Delta y}{\Delta x} = \frac{f(x+h) - f(x)}{x+h-x} = \frac{f(x+h) - f(x)}{h}$$

$$\tan \theta = \frac{\Delta y}{\Delta x}$$

$$\lim_{h \rightarrow 0} \frac{\Delta y}{\Delta x} = \frac{dy}{dx} = \text{slope} = f'(x) = \tan \theta = m = \left. \frac{dy}{dx} \right|_{(x, y)}$$

$$y = x^3$$

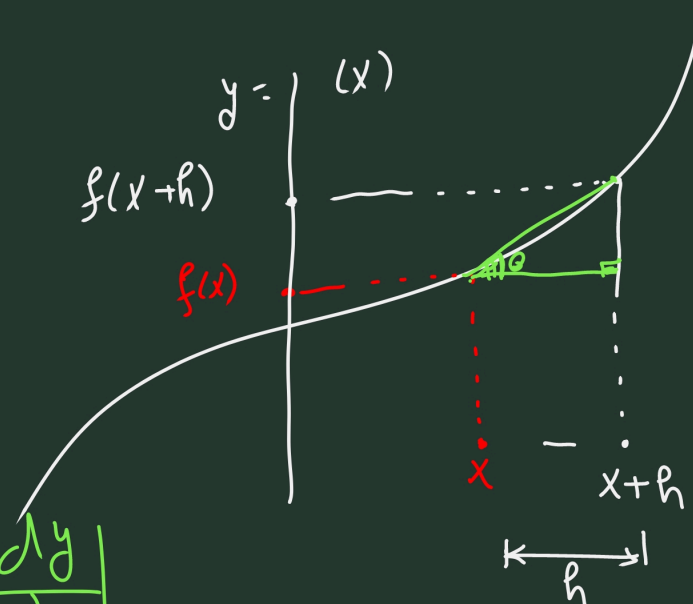
$$y' = 3x^2$$

$$y'' = 6x \quad y' = 0 \quad ?$$

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = \frac{0}{0}$$

$$\lim_{x \rightarrow 0} x \sin x = 0$$

$$\lim_{x \rightarrow 0} x \ln x = 0(\infty)$$



$$0 \cdot \infty =$$

L'Hopital Rule

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{0}{0}, \frac{\infty}{\infty}$$

$$\stackrel{L}{=} \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)} = \frac{0}{0}, \frac{\infty}{\infty}$$

$$\stackrel{L}{=} \lim_{x \rightarrow a} \frac{f''(x)}{g''(x)} = \dots = \textcircled{K}$$

$$y = \frac{f}{g}$$

$$y' = \frac{f'g - fg'}{g^2}$$

Case ①

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = \frac{0}{0}$$

$$\stackrel{L}{=} \lim_{x \rightarrow 0} \frac{\cos x}{1} = \frac{1}{1} = 1$$

Case ② $\infty - \infty$

→ cross-multiplication

Common factor

Geogebra

$$\lim_{x \rightarrow 0} \left(\frac{1}{x} - \frac{1}{\sin x} \right) = \frac{1}{0} - \frac{1}{\sin 0} = \infty - \infty$$

$$\stackrel{L}{=} \lim_{x \rightarrow 0} \frac{\sin x - x}{x \sin x} = \frac{0 - 0}{0 \cdot 0} = \frac{0}{0}$$

$$\stackrel{L}{=} \lim_{x \rightarrow 0} \frac{\cos x - 1}{x \cos x + \sin x} = \frac{1 - 1}{0 + 0} = \frac{0}{0}$$

$$\stackrel{L}{=} \lim_{x \rightarrow 0} \frac{-\sin x}{x \sin x + \cos x + \cos x}$$

$$= \frac{0}{0 + 1 + 1} = \frac{0}{2} = 0$$

$$y = \sin(x)$$

$$f = \frac{1}{T}$$

$$y = \sin(3x)$$

$$f = \frac{1}{T}$$

$$y = \sin\left(\frac{x}{3}\right)$$

$$\omega = 1$$

$$1 = 2\pi f$$

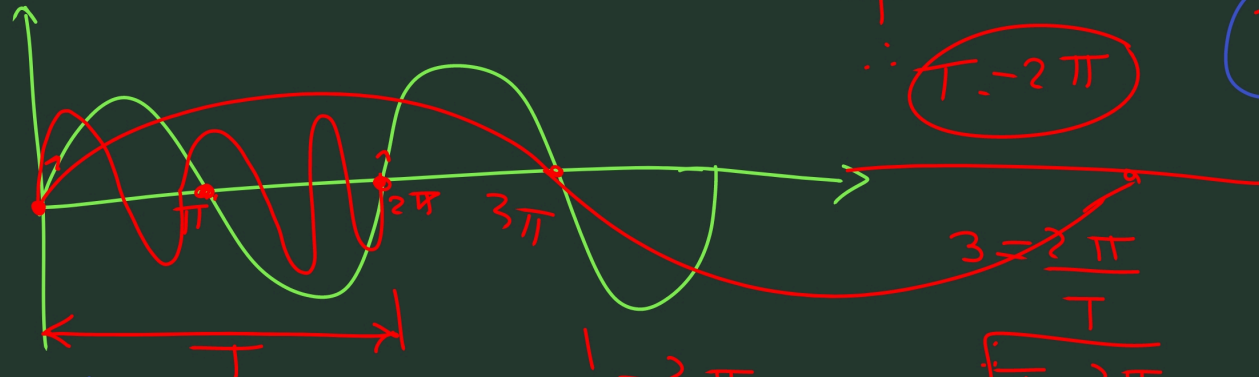
$$1 = \frac{2\pi}{T}$$

$$\therefore T = 2\pi$$

$$Z = 1 - (x^2 + y^2)$$

$$Z = x^2 + y^2$$

$$z = x^2 + y^2$$



$$\sin(3x) \quad y = \sin(\omega t) = \sin(t) \quad \omega = 1$$

$$\frac{1}{3} = \frac{2\pi}{T}$$

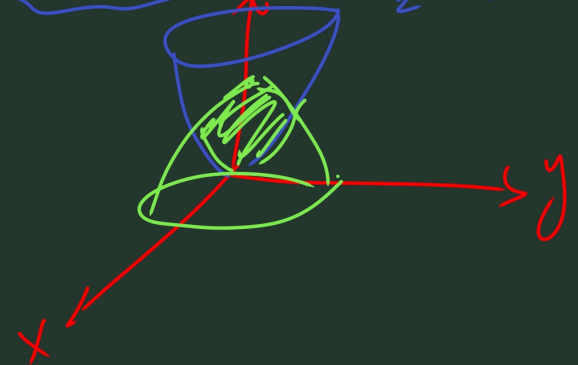
$$\omega = 2\pi f$$

$$\omega = \frac{2\pi}{T}$$

$$\therefore T = 6\pi$$

$$3 = \frac{2\pi}{T}$$

$$\therefore T = \frac{2\pi}{3}$$



Case 3 $0 \cdot \infty$
 $\downarrow \uparrow$

Case 4
 $\frac{\infty}{0} \neq \frac{0}{\infty}, \frac{\infty}{0}, \frac{0}{\infty}$

$$\lim_{x \rightarrow 0} x \ln x = 0 \cdot (-\infty) = - (0 \cdot \infty)$$

undefined value

$$= \lim_{x \rightarrow 0} \frac{\ln x}{\frac{1}{x}} = \frac{-\infty}{\frac{1}{0}} = \frac{\infty}{\infty}$$

$$\stackrel{L}{=} \lim_{x \rightarrow 0} \frac{\frac{1}{x}}{\frac{-1}{x^2}} = \lim_{x \rightarrow 0} -x = 0$$

$$y = \lim [f(x)]$$

$$y' = \frac{f'(x)}{f(x)}$$

Ex: $\lim_{x \rightarrow 0} (\tan x)^{\sin x} = (0)^0$

$$y = (\tan x)^{\sin x}$$

$$\ln y = \ln (\tan x)^{\sin x}$$

$$\boxed{\ln y} = \boxed{\sin x \ln (\tan x)}$$

$$\lim_{x \rightarrow 0} (\sin x) \ln (\tan x) = \lim_{x \rightarrow 0} \frac{\ln (\tan x)}{\frac{1}{\sin x}}$$

$$\textcircled{1} \ln(ab) = \ln a + \ln b$$

$$\textcircled{2} \ln\left(\frac{a}{b}\right) = \ln a - \ln b$$

$$\textcircled{3} \ln(a^n) = n \ln a$$

$$= 0 \cdot (-\infty) = - (0 \cdot \infty)$$

$$\lim_{x \rightarrow 0} \frac{\ln(\tan x)}{\csc x} = \frac{-\infty}{\infty}$$

$$= \lim_{x \rightarrow 0} \frac{\sec^2 x}{\tan x} \cdot \frac{1}{-\csc x \cdot \cot x}$$

$$= -\lim_{x \rightarrow 0} \frac{\sec^2 x}{\tan x \csc x \cot x}$$

$$= -\lim_{x \rightarrow 0} \frac{\cancel{\cos x} \sin x \sin x}{\cos^2 x \cdot \cancel{\sin x} \cancel{\cos x}}$$

$$= -\lim_{x \rightarrow 0} \frac{\sin x}{\cos^2 x} = \frac{0}{1} = 0$$

$$\ln y = \ln(\tan x)^{\sin x}$$

$$\ln y = 0$$

$$\cancel{\ln y} = 0$$

$$\cancel{y} = e^0$$

$$y = 1$$